Evaluation

Finding contradictions that span several steps of reasoning, and proving them. Measured on 549 documents from three external corpora and one purpose-built stress set. v0.8.8 · gpt-oss-120b · seed 7 · every figure reproducible from `tools/evaluate.py`.

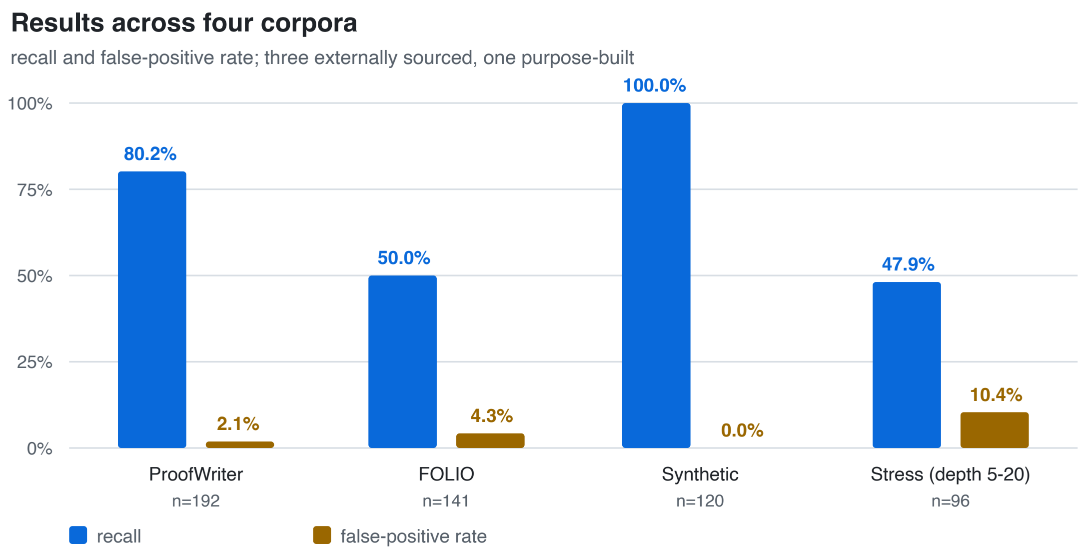
What is being measured

The system reads a document, translates each statement into first-order logic, and asks a solver whether the statements can all be true at once. When they cannot, it returns a **minimal conflicting set** and the derivation that reaches the conflict — not a verdict, but a proof that can be checked independently.

The evaluation asks two questions. Does it find real contradictions? And does it invent ones that are not there? The second matters more: a checker that reports spurious conflicts is worse than none.

Rates are reported with Wilson 95% confidence intervals. The intervals are not decoration — on 96 documents a single flip moves a rate by more than a point, and a reader entitled to ask "is that difference real?" needs them to answer.

Results

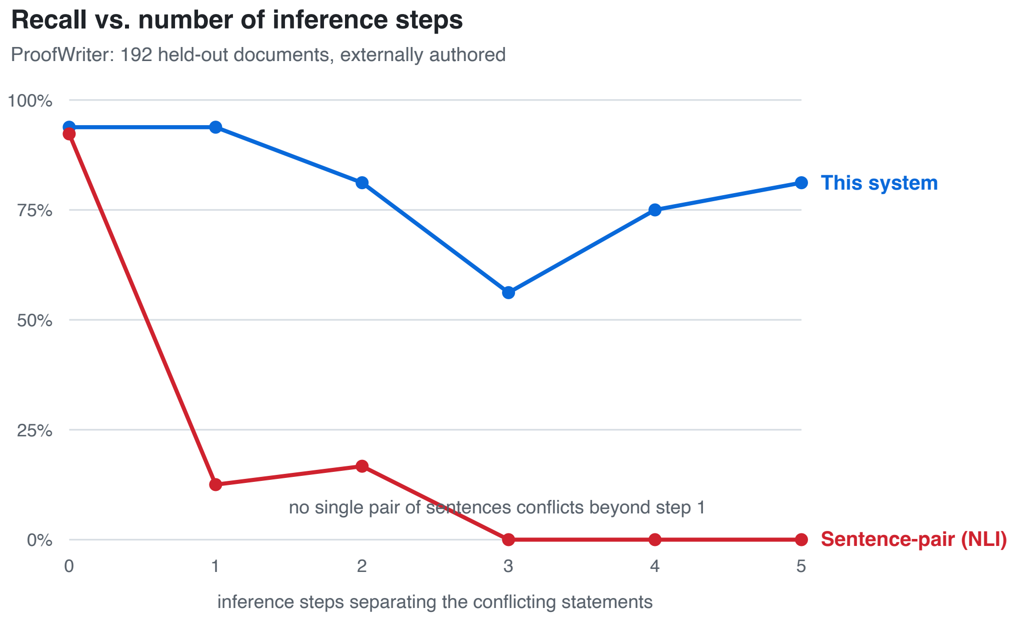


Corpus	n	Recall (95% CI)	False pos.	Precision	F1
ProofWriter	192	80.2% [71.1–87.0]	2.1%	97.5%	88.0
FOLIO	141	50.0% [38.8–61.3]	4.3%	92.3%	64.9
Synthetic	120	100% [94.0–100]	0.0%	100%	100
Stress (depth 5–20)	96	47.9% [34.5–61.7]	10.4%	82.1%	60.5

ProofWriter and FOLIO are externally authored and were not used during development. **Precision stays above 92% everywhere**, which is the property the design optimises for.

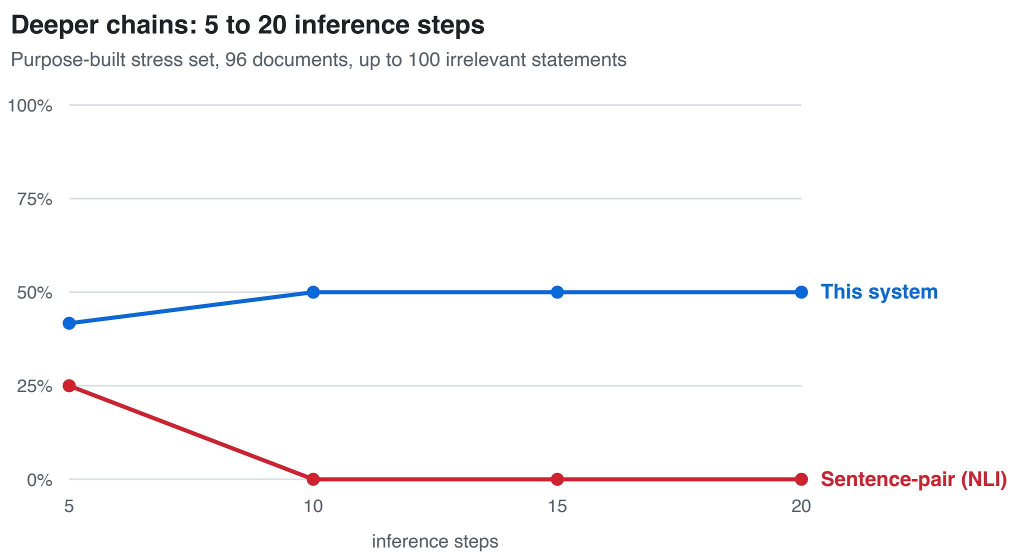
The multi-hop result

The standard approach to contradiction detection is sentence-pair entailment (NLI): compare statements two at a time and flag any conflicting pair. It has a structural limit. When a contradiction only emerges after chaining several statements together, **no individual pair of sentences conflicts**, and a method that only ever looks at pairs cannot see it.



Both systems ran on identical documents under identical scoring. At zero steps — where the contradiction is directly visible in one pair — they are level, 93.8% against 92.3%. **At one step the pairwise method collapses to 12.5% and by three steps it reaches 0%**, while this system stays above 56% through the deepest level the corpus contains.

Pushing the same axis further, on a purpose-built set reaching 20 steps with up to 100 irrelevant statements added as clutter:



Depth 5–20, 32 documents	Recall	False pos.	Calls per document
This system	**47.9%**	10.4%	~3 + solver
Sentence-pair (NLI)	6.2%	0.0%	**127.2**

The pairwise method reaches 25% at depth 5 and **0% at every deeper level**, while making 127 model calls per document — its cost grows with the square of document length, so the approach becomes more expensive exactly where it stops working. This system's recall is flat across depth, because a solver is indifferent to how long a chain is.

Where the limit actually is

Recall is flat at roughly 50% on the stress set *at every depth, including the shallowest*. That is not a reasoning failure — the solver handles 20 steps as easily as 5. It means a document is lost when **any single statement fails to translate**, since one missing link breaks a chain of any length.

Withholding the language model and translating with hand-written rules only, changing nothing else, isolates this:

	Full pipeline	Rules only
ProofWriter	80.2%	**12.5%**
Synthetic	100%	**1.7%**

Precision stays above 92% even with rules alone — deterministic translation is **sound but narrow**, yielding usable logic for a small fraction of real sentences. Translation, not solving, is the bottleneck. It is measured here two independent ways, and it is where further work belongs.

Three models were compared on ProofWriter under the same conditions (gpt-oss-120b 80.2%, DeepSeek-V4-Flash 70.8%, GLM-4.7 63.5%); the confidence intervals overlap, so the ordering is suggestive rather than established.

Limitations

- **Recall is bounded by translation coverage, not by the solver.** Roughly half the stress documents lose at least one statement in translation.
- **Only first-order structure is representable.** Contradictions resting on arithmetic, dates, degree or vagueness fall outside the fragment by construction, and no choice of model changes that.
- **Directly prompting a strong language model to judge a whole document is a serious alternative and is not evaluated here.** The comparison in this report is against sentence-pair entailment specifically. A language model reading the whole document does not face the pairwise structural limit, and any claim about relative detection accuracy against that approach would need its own experiment.
- **What the system offers over any prompting approach is a checkable artefact** — a minimal conflicting set and a derivation — rather than an unverifiable judgement. This report measures detection; it does not attempt to measure the value of that artefact.
- Confidence intervals on the stress set are wide (n=96, 48 positive).

Corpora: ProofWriter (Allen Institute for AI, CC BY 4.0) · FOLIO (Yale LILY, CC BY-SA 4.0) · synthetic and stress sets generated by `tools/make_stress_set.py` with recorded seeds. Reproduce with `python -m tools.evaluate --set validation/proofwriter`.